Directional Navigability Estimation from Ultra-Low-Resolution Multi-View Thermal Sensing

Anonymous Author(s)
Affiliation withheld for double-blind review

***Abstract*—Reliable perception of navigable indoor spaces remains a challenge for wearable navigation systems. Existing approaches typically rely on high-resolution cameras, depth sensors, or large annotated datasets, limiting their deployability in privacy-sensitive environments and increasing development costs. This paper investigates whether directional navigability can be estimated using only ultra-low-resolution thermal imagery from a wearable multi-camera platform. Training such systems is challenging due to the limited visual information available in low-resolution thermal sensors and the cost of collecting and annotating large thermal datasets. We propose a calibration-free learning framework that leverages high-resolution RGB imagery during data collection to automatically generate navigability labels, enabling the training of thermal perception models without manual annotation. The resulting models estimate navigability in multiple directions directly from multi-view thermal observations. Unlike conventional cross-modal approaches, the proposed framework does not require geometric calibration, pixel-level alignment, or fixed sensor configurations. Experiments using a wearable platform equipped with three ultra-low-resolution thermal cameras demonstrate that directional navigability can be reliably inferred from thermal sensing alone. We further compare alternative multi-camera fusion strategies and evaluate robustness to changes in camera arrangement. The results highlight the potential of ultra-low-resolution thermal sensing as a scalable and privacy-preserving solution for wearable indoor navigation.**

I. Introduction

Wearable navigation systems for visually impaired users have relied on RGB cameras or depth sensors that capture identifiable imagery, on large hand-labeled datasets, and on a fixed calibrated camera layout. A sensor that moves on the rig breaks the usual geometric assumptions.

Ultra-low-resolution thermal cameras are cheaper and more private. A 32×24 array costs on the order of tens of dollars, draws little power, and is too coarse to identify faces. Each tile also carries very little spatial detail, several cameras are needed for useful coverage, and thermal labels are slow to collect by hand. We test whether directional navigability (whether the left, middle, and right views are walkable or obstructed) can be estimated from thermal tiles alone, without calibration and without manual labels.

RGB is used only while collecting data (Fig. 1). A high-resolution camera on the wearable rig records frames in sync with three ultra-low-resolution thermal cameras. A pretrained detector on the RGB stream assigns a navigability label to each direction. At inference only the thermal tiles are available, and the model outputs a three-class state per direction (clear, person, other object). Supervision comes through discrete labels, not aligned pixels, so inference does not require calibration or a fixed camera-to-slot mapping.

We compare pixel-level stitching, decision-level output fusion, and feature-level fusion on the three thermal views, and we test slot permutations with and without a small identity classifier that re-sorts tiles before inference. The paper reports (i) an extension of RGB-supervised auto-labeling, following Gao *et al.*'s DareDevil [1], to a calibration-free multi-camera thermal array with no manual thermal labels and no RGB at inference; (ii) an empirical comparison of the three fusion strategies at this resolution; and (iii) an identity correction that recovers oracle-level accuracy after slot shuffles.

II. Related Work

A. Privacy-Preserving and Ultra-Low-Resolution Thermal Sensing

Wearable indoor navigation systems for visually impaired users have traditionally relied on RGB cameras or depth sensors, which raises privacy concerns when deployed continuously indoors. Gao *et al.*'s DareDevil [1] addresses this by auto-labeling thermal frames from a paired RGB camera through a pretrained YOLOv8 detector, avoiding manual annotation. However, DareDevil collapses all object classes into a single binary obstacle label and operates on colorized thermal frames from a single standard-resolution thermal camera at inference time; it does not address the extreme spatial compression of a 32×24-pixel array, multi-camera fusion, or robustness to sensor placement.

A separate line of work uses ultra-low-resolution thermal sensors (e.g., 24×32 arrays, on the order of \$30 per unit) as a low-power, lighting-invariant alternative to RGB and depth sensing. Such sensors are too coarse to resolve identifiable facial features, but their reduced spatial detail also degrades the features downstream perception models rely on; prior work compensates for this by fusing thermal imagery with gyroscope readings for motion estimation [5]. Naser *et al.* [4] separately showed that most low-resolution thermal pipelines still assume a fixed, calibrated sensor geometry. This work targets discrete navigability classification rather than continuous odometry, and removes the dependence on geometric calibration that these systems retain.

B. Automatic Label Generation for Traversability

Manually annotating walkable regions is expensive and does not scale to real-world indoor environments. Kim *et al.* [2]

[PLACEHOLDER]

Full-width pipeline figure.

End-to-end overview: wearable three-camera ultra-low-resolution thermal capture (RGB camera used only at data-collection time) → RGB/YOLO automatic per-direction label generation → the four thermal fusion models (stitched panorama, output fusion, feature fusion, feature fusion with camera-identity correction) → per-direction navigability predictions.

Fig. 1. End-to-end pipeline, from multi-view thermal data collection and RGB-based automatic labeling to the family of thermal fusion models evaluated in this paper. The RGB camera is present only during data collection; at inference the models use the three ultra-low-resolution thermal tiles alone. Thermal tiles are rendered here with the perceptually uniform "inferno" colormap for human readability only; the models themselves consume raw, single-channel intensity values with no colorization step.

addressed this for egocentric video with an automated annotation strategy that derives traversability labels without manual pixel-level annotation. We apply this same philosophy to a cross-modal setting: RGB-derived object detections supervise a thermal-only model only during data collection, and the RGB sensor is entirely absent at deployment.

C. Large-Scale and Web-Sourced Navigation Learning

CityWalker [3] trains an embodied urban navigation policy on over 2,000 hours of in-the-wild web video, using visual odometry to generate pseudo-labels instead of teleoperated demonstrations. CityWalker targets continuous waypoint-following from RGB video; this work targets discrete per-direction navigability from privacy-preserving thermal sensing, with an order of magnitude less data collected from a purpose-built wearable rig.

D. Multi-View Sensor Fusion and Placement Robustness

Multi-camera perception systems typically assume a fixed mapping between physical sensor position and input channel, an assumption that breaks when sensors shift or are reattached in a different order on a wearable rig. Pixel-level, decision-level, and feature-level fusion are well studied in the sensor fusion literature, but little work examines how these strategies behave when the camera-to-slot assignment is perturbed, particularly at the small camera counts and extreme resolutions considered here. This also relates to cross-resolution knowledge transfer, where a high-resolution signal supervises a model on coarser inputs [6]; our gap is larger still, since the supervisory RGB detector and the deployed 32×24 thermal arrays share no feature space and no geometric alignment at inference time. We contribute a direct comparison of three fusion strategies for an ultra-low-resolution, three-camera array, together with an explicit treatment of camera-slot permutation robustness.

III. Methodology

A. Problem Formulation

At each timestep the rig outputs three thermal tiles $\{x_{\mathrm{L}}, x_{\mathrm{M}}, x_{\mathrm{R}}\}$, one from each active camera, each 32×24. For each direction $d \in \{\text{left}, \text{middle}, \text{right}\}$ the model predicts $y_d \in \{\texttt{no_object}, \texttt{person}, \texttt{other_object}\}$. The class `no_object` means the direction is clear; the other two mark an obstruction, split by whether it is a person or another object. At inference the model reads only the thermal tiles: $f : (x_{\mathrm{L}}, x_{\mathrm{M}}, x_{\mathrm{R}}) \mapsto (y_{\mathrm{L}}, y_{\mathrm{M}}, y_{\mathrm{R}})$. Training labels come from the paired RGB stream (Section III-B). A per-direction validity mask drops timesteps where RGB labeling fails.

B. RGB-Based Automatic Label Generation

We follow Gao *et al.*'s DareDevil [1] in using a paired RGB camera and a pretrained detector to auto-label thermal frames without manual annotation. The extension here is calibration-free multi-camera labeling: instead of a single fixed RGB-to-thermal pixel alignment, each detection is routed to a direction by bounding-box overlap with a per-camera image footprint (below), which tolerates the three independent RGB-to-thermal homographies on this rig without requiring them to be pixel-accurate.

Labels are generated from synchronized RGB frames with no manual step. Each frame is passed through a pretrained YOLOv8 detector. COCO class `person` maps to `person`; every other detection maps to `other_object`. To assign a detection to a direction, its bounding box is checked for overlap with the image footprint of each thermal camera, using RGB-to-thermal homographies fixed when the rig is built. If a person box overlaps a camera footprint, that direction is `person`. If only a non-person box overlaps, it is `other_object`. With no overlap, it is `no_object`. Directions without a valid footprint are masked out. The thermal model never sees RGB at inference and is not trained on aligned RGB pixels.

C. Thermal Fusion Architectures

We evaluate four architectures for estimating per-direction navigability from the three active thermal cameras (left, middle, right), each consuming only raw 32×24 thermal tiles at inference time and producing a 3-class prediction per camera over $\{\texttt{no_object}, \texttt{person}, \texttt{other_object}\}$. Tiles are kept as single-channel raw intensity throughout, with no colorization step converting them to pseudo-RGB before the backbone, unlike prior work that colorizes thermal frames for a detector pretrained on natural images. The four span pixel-level, decision-level, and feature-level fusion, plus a fourth

variant that adds a learned correction for camera-placement permutations on top of feature-level fusion. All but the stitched-panorama baseline share a configurable convolutional backbone, `FlexBackbone` (depth $d \in \{2, \ldots, 5\}$, width multiplier w, activation $a \in \{\text{ReLU}, \text{GELU}, \text{LeakyReLU}\}$), which returns the intermediate feature map at every block so downstream heads can pool over multiple resolutions.

a) Model 1: Stitched Panorama (pixel-level fusion): The three raw tiles are warped into a shared panoramic canvas via pre-computed thermal-to-thermal homographies, using nearest-neighbor interpolation so warped pixels keep their original raw thermal values rather than being blended with neighbors or with the zero-padded border, producing a two-channel panorama (raw intensity plus a validity mask). A fixed five-layer backbone is applied independently to each camera's bounding-box crop of the panorama, followed by a shared classification head. Because the panorama is built once, before any learned processing, cross-camera information lost to the homography warp is unavailable to the model; this baseline isolates the cost of fusing at the pixel level.

b) Model 2: Output Fusion (decision-level fusion): A shared-weight `FlexBackbone` processes each camera's tile independently, and a per-camera head reduces it to one 3-class logit vector. The three logit vectors are concatenated and passed through a small fusion MLP that refines them jointly. Fusion only ever sees already-classified, low-dimensional logits, so cross-camera interaction is limited to statistical dependencies between directions rather than raw spatial features.

c) Model 3: Feature Fusion: The same shared-weight backbone runs on each camera's tile, but instead of classifying each camera separately, the feature maps from all three cameras at a given backbone level are concatenated channel-wise and projected back to the original channel count via a learned 1×1 convolution, applied at every level. This yields a single fused multi-scale feature pyramid mixing all three cameras at every depth, in contrast to Model 2's late fusion. A shared head then produces the final per-camera logits. We sweep backbone depth, width, activation, and head capacity to select the configuration used as the reference Feature Fusion result throughout this paper.

d) Model 4: Feature Fusion with camera-identity correction: Model 3's per-level 1×1 projections consume all cameras' features in a fixed slot order, so the projection layer can learn to associate an input slot with a physical direction even though the backbone itself is fully shared and permutation-agnostic; if a sensor is reattached in a different slot at deployment time, this learned positional dependence can degrade predictions. We correct for it with a lightweight auxiliary `CameraIdentityClassifier`, a compact CNN that takes one raw thermal tile and predicts a 3-way softmax over $\{\text{left}, \text{middle}, \text{right}\}$, trained independently using the true camera-to-direction mapping recorded at capture time; on its own held-out validation split it reaches 98.7% accuracy on this 3-way slot-identification task, against a 33.3% random-guess baseline. At inference, given a possibly shuffled set of tiles, the classifier scores every slot against every direction and picks the assignment that maximizes total log-probability, enumerated exhaustively over all $3! = 6$ permutations; the resorted tiles are then passed to the unmodified Model 3. We compare this against an oracle re-sort using ground-truth identities (an upper bound on achievable robustness) and a blind condition with no correction at all.

All four architectures share the same masked, per-camera, three-class cross-entropy training objective, gated by a per-camera validity mask so that timesteps without a labeled RGB frame do not contribute gradient.

IV. Experimental Setup

A. Wearable Thermal Data Collection Platform

We developed a wearable thermal sensing platform designed for privacy-preserving indoor navigation research. The platform consists of multiple ultra-low-resolution thermal cameras mounted on a wearable frame and connected to an embedded computing unit for synchronized data acquisition. The design emphasizes low cost, privacy preservation, and flexibility, making it suitable for studying navigation perception under severe sensing constraints.

The wearable platform incorporates multiple MLX90640 SparkFun IR Array Breakout sensors, each providing a 32×24 thermal image of the environment. The sensors are mounted at different viewing angles to increase environmental coverage and capture complementary observations of the user's surroundings. All sensors are connected to an NVIDIA Jetson Nano-based acquisition unit that records timestamped thermal frames for subsequent processing.

B. Data Collection

C. Training Protocol

V. Results

A. Navigability Estimation Performance

Figure 2 compares the four architectures with per-class F_1 per camera direction. Every model beats chance on the raw 32×24 tiles. On validation, Feature Fusion leads the non-shuffled models with overall per-class F_1 around 0.81 (clear), 0.83 (person), and 0.80 (other object). The stitched panorama and output-fusion baselines sit near 0.73–0.78 overall and are weakest on person detection from the right camera, where the thermal signal is thinnest.

B. Fusion Strategy Analysis

Feature-level fusion beats pixel-level stitching and decision-level output fusion. Concatenating feature maps across cameras at each backbone level keeps spatial cues that stitching loses in the warp and that output fusion never sees because it only merges logits. Feature fusion runs more backbone passes, but on tiles this small the gain is large enough to use it in the shuffle experiments below. On a single-sample CPU forward pass, Baseline 1, Baseline 2, and Feature Fusion take 1.36, 0.80, and 2.37 ms respectively, and the camera-identity classifier adds another 0.21 ms; every configuration stays under 3 ms, well within budget for real-time inference on a wearable device.

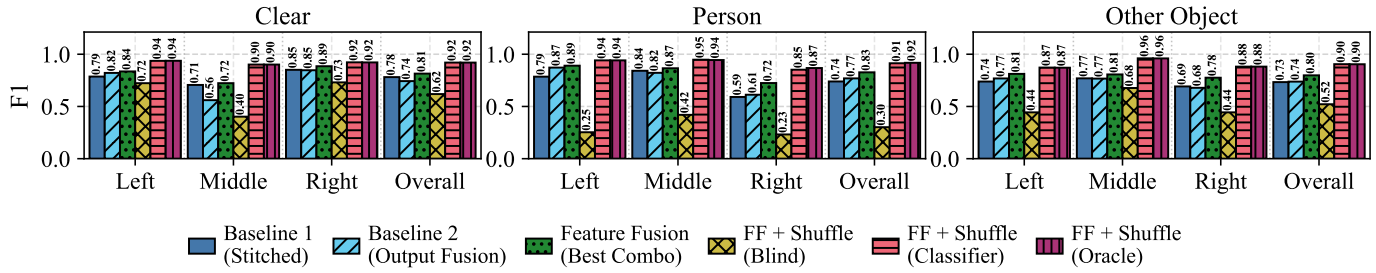


Fig. 2. Per-class F_1 by camera direction for the obstruction-detection architectures. Baselines and Feature Fusion are evaluated on the validation set; Feature Fusion with shuffle correction (blind, classifier, oracle) is evaluated on the held-out test set.

C. Camera Arrangement Robustness

Figure 3 shows Feature Fusion on permuted camera slots in the held-out test set. With no correction (blind), shuffling slots drops overall per-class F_1 to about 0.40–0.42: the fixed-order projection layers in Model 3 encode slot position. A small camera-identity classifier re-sorts the tiles before inference and brings overall F_1 back to about 0.90–0.92 on all three classes, within rounding of the oracle that uses ground-truth slot identities. The backbone features carry direction information; what breaks under shuffle is the slot ordering, not the thermal representation itself.

Data Availability

The thermal-RGB dataset collected on the wearable platform will be made publicly available upon publication.

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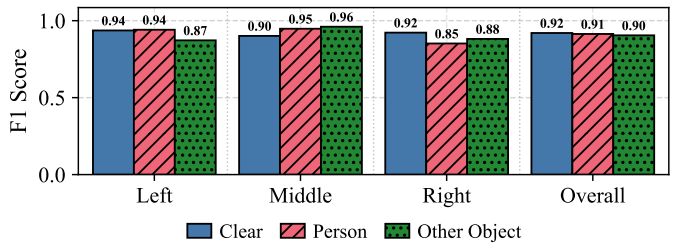


Fig. 3. Per-class F_1 of Feature Fusion with classifier-based shuffle correction on the held-out test set, broken down by camera direction (left, middle, right) and overall.